Table 1: **The reasoning gain is not tied to OpenThoughts3.** Change in MATH500 accuracy when each base's Tulu3 SFT checkpoint receives 30K samples from one of four reasoning corpora, on the common 287-problem subset left after excluding the identified overlaps (Section **??**). The corpora are distinct releases but overlap with MixtureVitae's pretraining pool, and the runs match examples and epochs, not supervised tokens. Brackets are 95% paired bootstrap intervals. **Bold** and *italic* mark significant increases and decreases after Benjamini–Hochberg correction over the 20 changes. Appendix Table **??** gives all 500 problems. Protocol B (Appendix Table **??**).

Base	**OpenThoughts3**	**AceReason**	**Nemotron-Cascade**	**Superior-Reasoning**
MixtureVitae	**+41.9** [+37.6, +46.5]	**+38.7** [+34.3, +43.1]	**+38.4** [+34.0, +42.5]	**+26.6** [+22.5, +30.8]
Comma0.1	−0.1 [−1.4, +1.2]	+0.5 [−0.8, +1.7]	+1.6 [+0.3, +3.0]	+0.5 [−0.8, +1.7]
FineWeb-Edu	+0.0 [−0.9, +0.8]	+1.0 [+0.0, +2.2]	+0.9 [−0.1, +2.0]	+1.3 [+0.1, +2.6]
Nemotron-CC-HQ	*−1.6* [−3.0, −0.5]	−0.6 [−1.9, +0.7]	−0.2 [−1.7, +1.3]	−0.8 [−1.9, +0.3]
SmolLM2	+0.3 [−2.2, +2.8]	**+7.2** [+4.3, +10.2]	**+7.2** [+4.2, +10.1]	**+7.5** [+4.8, +10.2]

Table 2: **Benchmark-overlap filtering before training, and the audit.** How each training dataset was filtered for benchmark overlap before training, and what the audit read. "Not audited" means that no item-level match was looked for.

Training data	Filtered before training	Audited
16K extension data (`long_sft`)	None	Every message
Tulu3 SFT mixture, as trained	MixtureVitae 13-gram filter	Every message
Tulu3 preference data, as trained	The same filter, probably on the prompts only	Prompt and both answers
Phase 2 30K samples of the four reasoning corpora	Not recorded	As trained, after the 16K cut
OpenThoughts3 source of MixtureVitae	Not recorded	Its 65,089 questions
MixtureVitae's other reasoning data (37 shards)	Filtered, index not established	Every document
Rest of MixtureVitae pretraining (web and other text)	Filtered, index not established	Not audited
Reference pretraining datasets and external models	As released	Not audited

Table 3: **Benchmark items identified in the training data.** Items is the number scored, Found the items identified anywhere, and Left those kept in the comparisons without them. The last six columns count the items found in each training dataset: the 16K extension data, the Tulu3 SFT and preference data as trained, the Phase 2 30K samples as trained, the OpenThoughts3 questions and the public samples of the three other reasoning corpora, and MixtureVitae's pretraining reasoning pool other than OpenThoughts3. An item can be found in several. No item of AIME25, HumanEval, GPQA-Diamond, IFEval or LAMBADA was identified, which means no item-level match under this procedure, not absence.

Benchmark	Items	Found	Left	16K ext.	Tulu3 SFT	Tulu3 DPO	Phase 2 30K	Reasoning questions	MV pool
CommonsenseQA	1,221	670	551	15	650	204	12	4	0
MATH500	500	213	287	2	1	0	7	1	211
LiveCodeBench	369	199	170	0	0	0	0	0	199
JEEBench	515	172	343	0	0	0	3	1	169
OpenBookQA	500	162	338	0	0	0	0	0	162
MMLU	14,042	79	13,963	9	13	3	13	0	62
COPA	100	45	55	0	2	7	0	0	39
ARC-Easy	2,376	36	2,340	1	23	11	0	0	3
AMC23	40	26	14	0	0	0	2	1	26
AIME24	30	25	5	0	0	0	0	0	25
BoolQ	3,270	21	3,249	0	15	3	0	4	0
MBPP	500	18	482	0	1	0	3	1	15
WinoGrande	1,267	17	1,250	0	14	6	0	0	0
PIQA	1,838	14	1,824	0	4	3	0	0	10
ARC-Challenge	1,172	12	1,160	0	8	3	0	0	1
HellaSwag	10,042	10	10,032	0	8	1	0	0	1
GSM8K	1,319	1	1,318	0	0	0	0	0	1
All		1,720							

Table 4: **The principal comparisons with and without the identified items.** Changes in points with 95% intervals, each from its own analysis (Sections ?? and ??). Without the items, MATH500 keeps 287 of its 500 problems, and every compared checkpoint is scored on the same items. The suite averages keep the LiveCodeBench and MBPP items found, because the Phase 2 records hold no per-question verdicts for the code tasks.

Comparison	All items	Without the identified items
MV SFT→OT30K, MATH500	+41.8 [+38.4, +45.2]	+41.9 [+37.6, +46.6]
MV SFT→OT30K, suite average	+10.7 [+8.2, +13.2]	+11.0 [+8.1, +13.9]
OT3-source exclusion, difference of MATH500 gains	+14.5 [+10.9, +18.4]	+11.5 [+6.3, +16.8]
OT3-source exclusion, difference of suite-average gains	+5.9 [+2.2, +9.9]	+7.8 [+3.5, +12.1]
MV SFT→OT30K, language average	+0.23 [−0.23, +0.66]	+0.20 [−0.21, +0.68]

Table 5: **Every comparison without the identified items.** For each family of comparisons: the number of comparisons, the largest change of an estimate in points, and how many cross the significance threshold ($q < 0.05$) in either direction. Leaving out many of a benchmark's items widens its intervals, so a call can change with little change in the estimate. Table 6 lists the calls that change, and the released audit has every comparison.

Comparisons	n	Largest change	Calls changed
Table 2, changes of the suite average	11	1.39	0
Table 2, MATH500 changes	11	2.29	1
OT3-source exclusion, gains and their difference	6	3.04	0
Table 4, MATH500 changes	20	1.93	1
Factorial, front-loading effects	24	1.84	0
Factorial, source-pool contrasts	24	1.61	5
Factorial, interactions	12	1.02	2
Language, changes of the average	29	0.42	8
Language, front-loading contrasts of the average	20	0.46	1
Language, changes of single tasks	290	5.64	7
Language, front-loading contrasts of single tasks	200	5.82	11

Table 6: **Comparisons whose significance changes without the identified items.** Estimates in points with 95% intervals and Benjamini–Hochberg q-values, on all items and without the identified ones, for the families of Table 5 other than single language tasks.

	All items		Without them	
Comparison	Estimate	q	Estimate	q
MV DPO→OT30K, MATH500	+4.3 [+1.3, +7.2]	0.007	+2.0 [−1.7, +5.9]	0.406
FineWeb-Edu SFT→AceReason, MATH500	+1.1 [+0.3, +1.9]	0.006	+1.0 [+0.0, +2.2]	0.096
Source pools, MV − Nemotron-CC-HQ, 4K, SFT	+0.8 [−1.0, +2.5]	0.404	+2.4 [+0.2, +4.6]	0.038
Source pools, MV − FineWeb-Edu, 4K, SFT	+0.7 [−0.9, +2.3]	0.415	+2.3 [+0.3, +4.2]	0.028
Source pools, MV − FineWeb-Edu, 4K, DPO	+1.3 [−0.6, +3.1]	0.223	+2.8 [+0.7, +5.2]	0.013
Source pools, MV − Nemotron-CC-HQ, 16K, 16K base	+2.6 [+0.8, +4.3]	0.009	+2.2 [−0.1, +4.5]	0.072
Source pools, MV − FineWeb-Edu, 16K, 16K base	+3.0 [+1.1, +4.9]	0.005	+3.1 [+0.0, +6.4]	0.065
Interaction, MV − non-permissive, 4K, SFT	−2.7 [−4.6, −1.1]	0.003	−1.8 [−4.4, +0.6]	0.526
Interaction, MV − non-permissive, 4K, DPO	−2.7 [−4.6, −0.7]	0.034	−1.9 [−4.5, +0.9]	0.526
MV 16K base, from 4K base, language	−0.87 [−1.56, −0.15]	0.042	−0.80 [−1.71, +0.19]	0.140
MV OT30K, from 16K base, language	+0.69 [+0.30, +1.10]	0.004	+0.75 [+0.18, +1.34]	0.059
MV SFT→OT30K, from 16K base, language	+0.72 [+0.12, +1.35]	0.042	+0.86 [+0.04, +1.63]	0.109
MV DPO→OT30K, from 16K base, language	+0.78 [+0.19, +1.41]	0.042	+0.86 [+0.05, +1.63]	0.109
Comma0.1 SFT, from 16K base, language	−0.95 [−1.58, −0.29]	0.005	−0.85 [−1.70, +0.04]	0.081
Comma0.1 DPO, from 16K base, language	−0.75 [−1.37, −0.13]	0.018	−0.73 [−1.60, +0.15]	0.116
Nemotron-CC-HQ SFT→OT30K, from SFT, language	−0.53 [−1.00, −0.10]	0.045	−0.31 [−0.83, +0.15]	0.431
FineWeb-Edu 4K DPO, from 4K SFT, language	+0.09 [+0.03, +0.15]	0.012	+0.10 [+0.02, +0.19]	0.060
Interaction, MV − non-permissive, 16K base, language	+1.22 [+0.15, +2.33]	0.028	+0.76 [−0.78, +2.27]	0.299

Table 7: **The Phase 2 samples' questions in MixtureVitae's reasoning pool.** Percent of the questions of each 30K sample trained on that match a document of MixtureVitae's pretraining reasoning data other than OpenThoughts3, overall and by domain (13-gram and exact matching, Appendix **??**). Other is general questions for Nemotron-Cascade and instruction following for Superior-Reasoning. The model pretrained without the OpenThoughts3 source keeps this data, so the OT3-source exclusion keeps most of the Phase 2 OpenThoughts3 mathematics problems, with other solutions. A match is a document of the pool, not evidence that a 300B-token run drew it.

Corpus	Questions	All	Math	Code	Science	Other
OpenThoughts3	30,000	80.6	97.8	51.0	5.8	–
AceReason	30,000	66.4	65.8	67.7	–	–
Nemotron-Cascade	30,000	51.0	65.9	68.0	12.5	8.2
Superior-Reasoning	30,000	46.0	37.8	97.1	41.4	0.3

Table 8: **Training tokens of each reasoning dataset.** The 30K samples of the MixtureVitae runs as trained: LLaMA-Factory's tokenized caches, in the open-sci tokenizer, after the chat template and the cut at 16,384 tokens. At 16,384 is the percent of samples cut at the limit, whose target ends without the end-of-turn token, and Ends the percent trained to stop. Supervised tokens count the answer tokens over the five epochs. The runs match examples and epochs, not supervised tokens. In the OpenThoughts3 sample as released, 62.7% of the answers never close their reasoning, before any cut.

Dataset	Samples	Tokens per sample	At 16,384 (%)	Ends (%)	Supervised tokens
OpenThoughts3	30,000	14,861	62.4	37.6	2.20B
Superior-Reasoning	30,000	9,763	38.8	61.3	1.42B
AceReason	30,000	7,517	4.5	95.5	1.10B
Nemotron-Cascade	30,000	6,044	3.2	96.8	0.88B